

AMAS: Problem set 1



Aske Roelshøj (KVF851)
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Problem 1

For this assignment, the Ken Pomeroy data is loaded using Pandas, i.e. I download the website as a html and load the data as a dataframe.

On the website, the column-labels are repeated. I remove these, and convert the DRtg and ORtg columns to numerics - so I can manipulate them. When loading the data, beside team names there are sometimes numbers, so for later i strip the names of these numbers aswell.

Then I make a list of the conference names of interest, and retrieve DRtg values for these.

Putting the data in a histogram, we get:

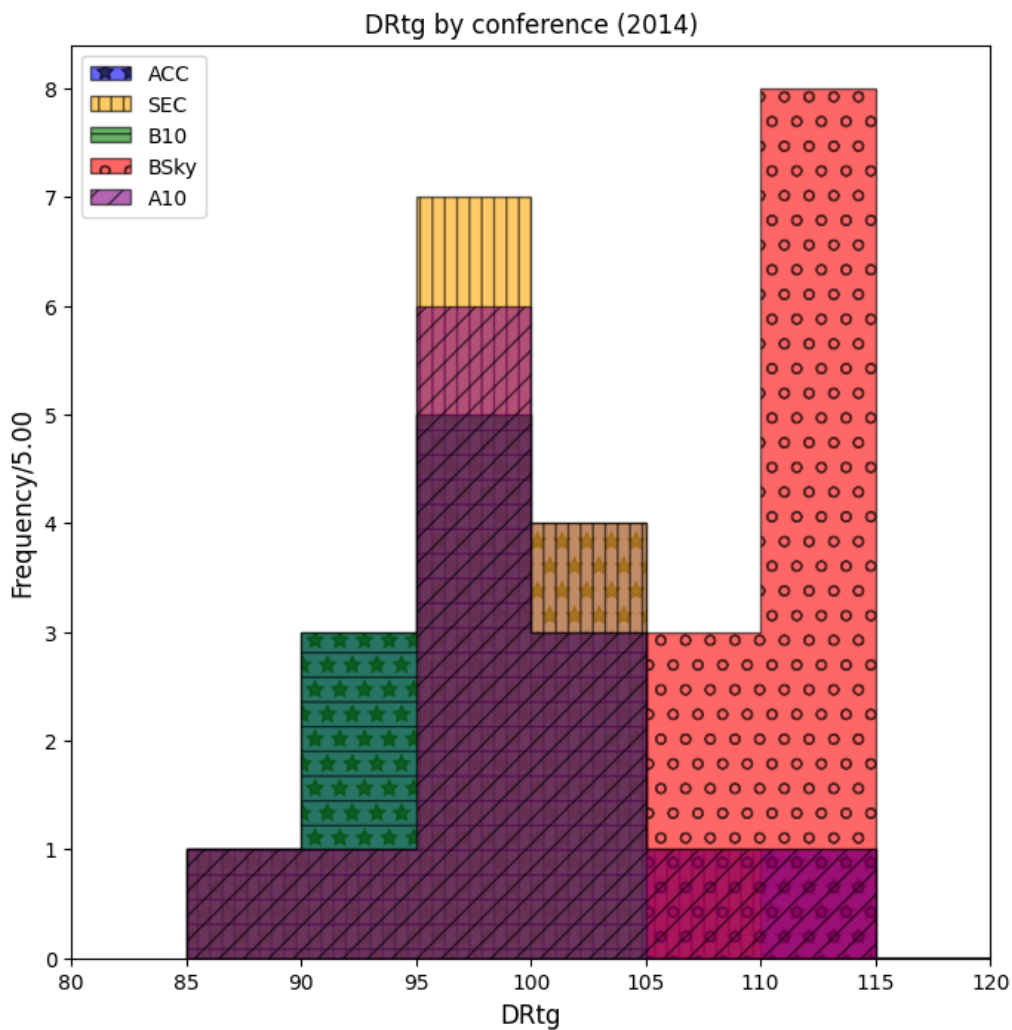


Figure 1: Histogram of DRtg scores across the 5 conferences

Problem 2

I now also load the 2009 Ken Pomeroy data, in the same way as before.

It is not guaranteed that the same teams that played in 2014 are also represented in 2009, so the difference in ORtg (Δ_{ORtg}) can only be found for "common" teams, i.e. the same teams that are represented in both years in the same conferences. Other data points are omitted.

The solution is to simply merge the two dataframes by teams and conferences - so we can find the difference in ORtg for the same conferences.

Plotting the difference in ORtg against ORtg values for 2009, we get:

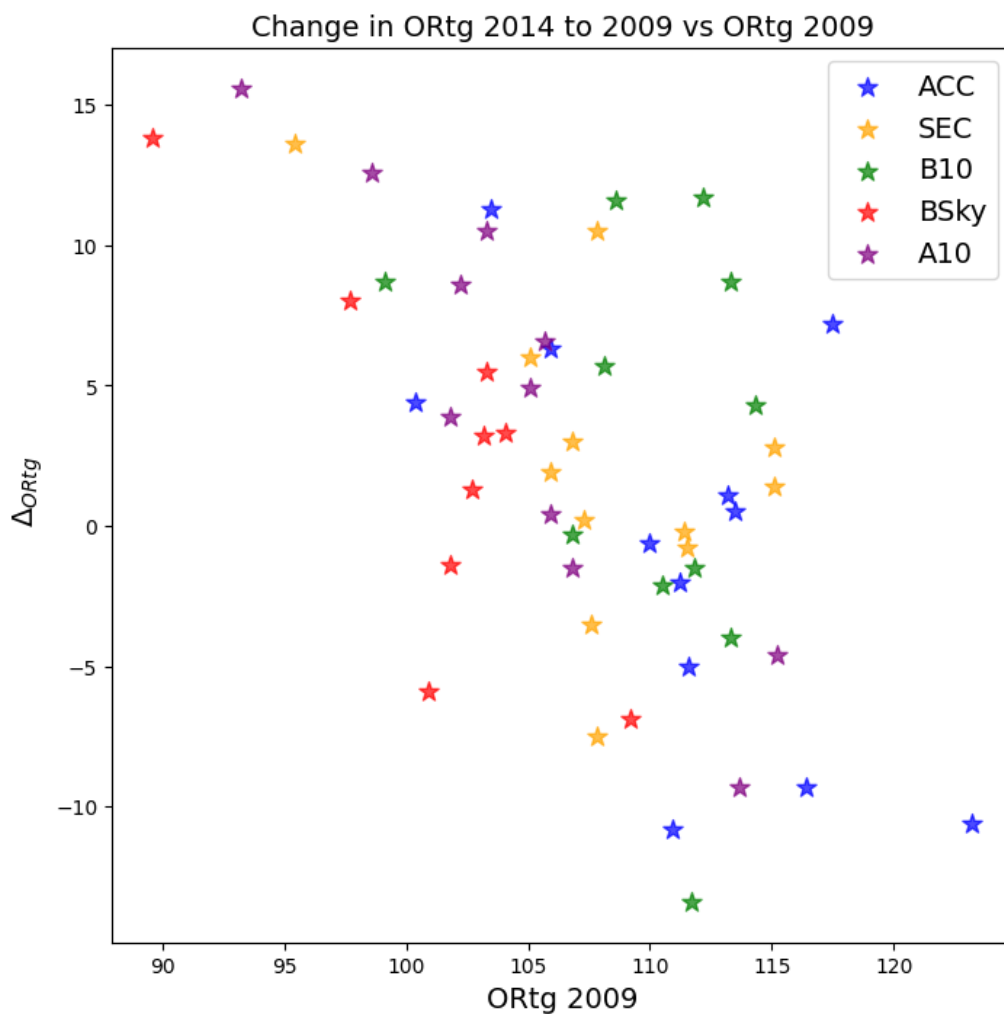


Figure 2: Δ_{ORtg} vs. ORtg values for 2009 across the same 5 conferences.

We calculate the mean and median Δ_{ORtg} for across the teams across each conference. For the teams not represented in the five conferences and who do not switch, the mean and median value is found in the same way. The results are represented in (table 1):

Conference	Mean	Median
<i>A10</i>	4.34	4.90
<i>ACC</i>	-0.63	-0.05
<i>B10</i>	2.67	4.30
<i>BSky</i>	2.32	3.20
<i>SEC</i>	2.28	1.65
<i>BE</i>	1.14	1.90
<i>Other</i>	2.40	1.30

Table 1: Table showcasing the mean and median values for Δ_{ORtg} across teams represented in the 5 conferences and others (these are the teams not in the 5 conferences).

Problem 3

Adding BE to the conference names, and redoing prob. 1 & 2, we get, first the following histogram on fig. 3.

We also get the following plot of Δ_{ORtg} on fig. 4.

And the following means and medians:

Conference	Mean	Median
<i>A10</i>	4.34	4.90
<i>ACC</i>	-0.63	-0.05
<i>B10</i>	2.67	4.30
<i>BSky</i>	2.32	3.20
<i>SEC</i>	2.28	1.65
<i>BE</i>	1.14	1.90
<i>Other</i>	2.48	1.50

Table 2: Table showcasing the mean and median values for Δ_{ORtg} across teams represented in the 5 conferences now with BE and others (these are the teams not in the 6 conferences).

Looking at table 2, I get at different mean and median for other conferences with BE than without. Upon comparing my results to my friends, they get slightly different values. I have though tried to debug my code dozens of times - but I still get the above results...

Note: In the notebook, I also calculate errors on means and medians (using jackknife resampling), but I ended up not using these, since we are simply calculating the mean and medians, and not trying to estimate a specific global parameter, and the data is not from a larger "parent" sample.

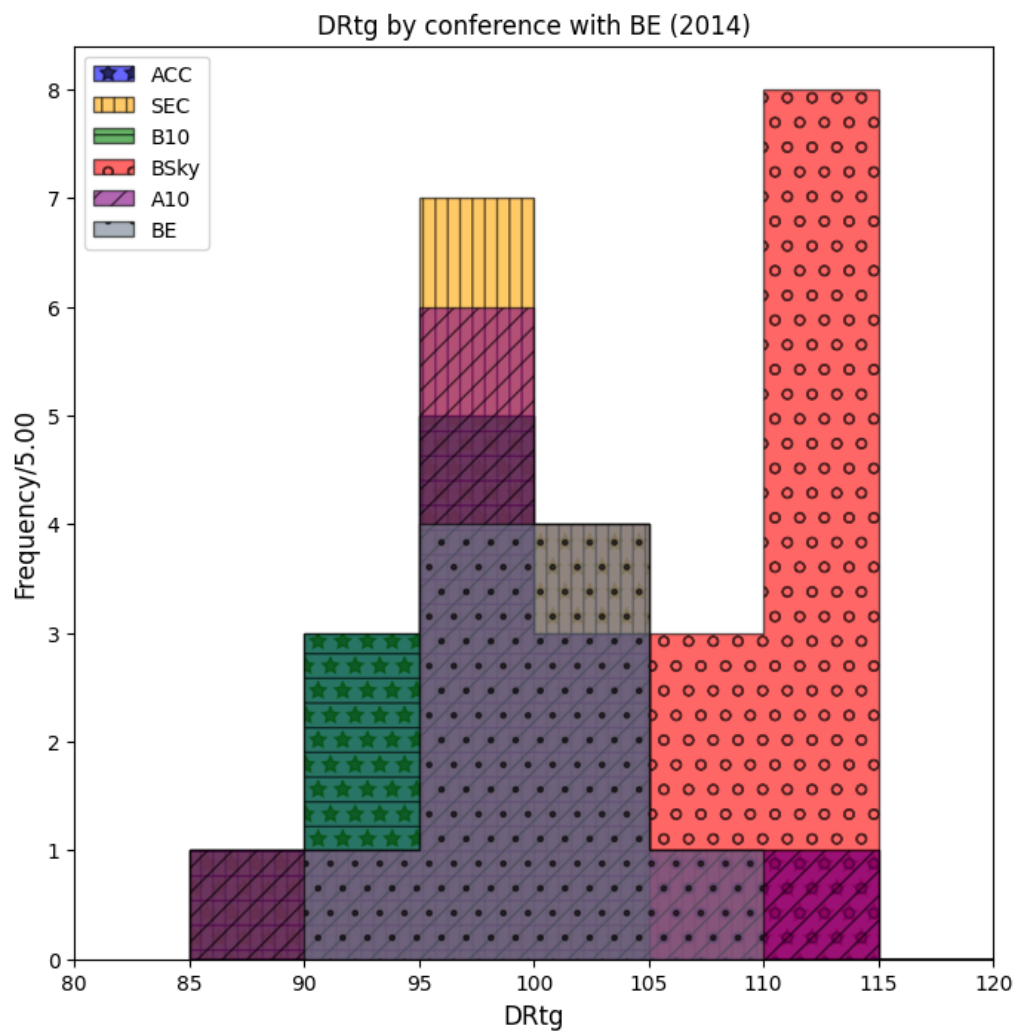


Figure 3: Histogram of DRtg scores across the 5 conferences now with BE added

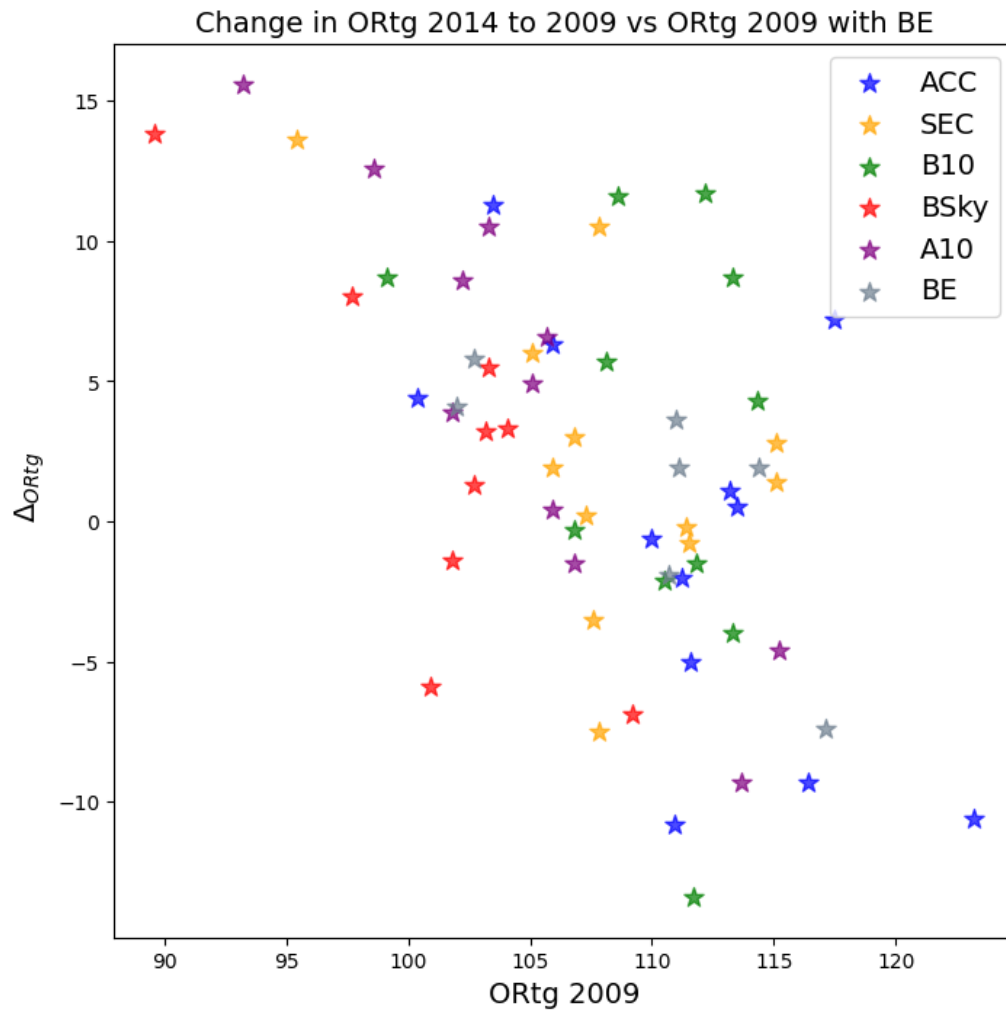


Figure 4: Δ_{ORtg} vs. ORtg values for 2009 across the same 5 conferences now with BE.

Extra Credit

This problem ended up being very(!) time consuming for me. For this problem I use the `pdfplumber` and its `extract_text` function to read the pdf and extract the text.

When looking at `AuthorsList.pdf`, we notice that the text comes in different fonts. We solved this problem using this fact. It is evident that the upper-half of the first page is part of the paper bibliography, i.e. we want to omit this part. Furthermore, we also want to omit the reference numbers, pagenumbers and headers.

Separating the text in terms of fontsize and printing the text, we see that there is a distinct fontsize reserved for authors (≈ 9.9626) and one for collaboration groups (≈ 10.4607), and others for the parts we do not care for - so we throw these away.

We then clean, i.e. we remove commas, the dataframe containing group names, and remove these from the author dataframe. The author dataframe is then cleaned by removing any pagenumbers, "and"s and spaces, such that the author names are in the same format, i.e. no spaces between name and last name, just periods separating them.

I get 3614 unique authors.

After this, we make a function which orders last name first and then first name(s), such that we can sort our dataframe in the correct way alphabetically. Since the amount of unique authors is even, there are two "midpoint" authors.

I get the midpoint authors to be K. Link and S. D. Linker.

The code is uploaded as a separate notebook.